

Practical 1 — Combined Redirection + Pipes

```
sairedy@DESKTOP-HKQRL9G: ~
GNU nano 7.2
Linux is an operating system.
Linux supports processes.
Windows is another operating system.
Linux uses pipes and redirection.
```

```
cd Combined_Redirection
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection$ nano input.txt
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection$ cat input.txt | grep Linux > output.txt
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection$ cat output.txt
Linux is an operating system.
Linux supports processes.
Linux uses pipes and redirection.
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection$ |
```

Practical 2 — Complex Commands

```
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection$ cd ~/Combined_Redirection
-bash: cd: /home/sairedy/Combined_Redirection: No such file or directory
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection$ cat count.txt
3
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection$ |
```

Practical 3 — Background Jobs

```
sairedy@DESKTOP-HKQRL9G: ~
GNU nano 7.2
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Background process started\n");

    for (int i = 1; i <= 10; i++)
    {
        printf("Running... %d\n", i);
        sleep(2);
    }

    printf("Background process completed\n");

    return 0;
}
```

```
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection/Background_Jobs$ ./task &
[1] 803
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection/Background_Jobs$ Background process started
Running... 1
Running... 2
Running... 3
Running... 4
Running... 5
Running... 6
Running... 7
Running... 8
Running... 9
Running... 10
Background process completed
[1]+ Done ./task
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection/Background_Jobs$ jobs
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection/Background_Jobs$ ps
  PID TTY          TT TIME CMD
 806 pts/0    00:00:00 ps
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection/Background_Jobs$ ps -f
  UID      PID  PPID  C  TIME TTY          TT TIME CMD
sairedy    318    0 09:31 pts/0    00:00:00 -bash
sairedy    807    318 09:33 pts/0    00:00:00 ps -f
sairedy@DESKTOP-HKQRL9G:~/MultiPipe_Practical/Input_Redirection/Combined_Redirection/Background_Jobs$ top
top - 09:34:02 up 32 min, 1 user, load average: 0.00, 0.02, 0.00
Tasks: 22 total,  1 running, 21 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 3758.6 total, 3192.7 free,  438.0 used, 282.7 buff/cache
MiB Swap: 1624.0 total, 1624.0 free,  0.0 used, 3320.7 avail Mem

  PID USER      PR  NI  TSTIME CPU    %CPU  %MEM    TIME+ COMMAND
    1 root        20   0   09:34   0.0    0.0   0.1   0:01.54 systemd
  318 root        20   0   09:34  0.0    0.0   0.1   0:00.00 -bash
  807 sairedy     20   0   09:34  0.0    0.0   0.1   0:00.00 ps -f
```

```
saireddy@DESKTOP-HKQRL9: + -
top - 09:33:14 up 31 min, 1 user, load average: 0.00, 0.02, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.7 sy, 0.0 ni, 99.3 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3758.6 total, 3191.1 free, 439.7 used, 202.6 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used, 3318.9 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   21676   13060   9656  S   0.0   0.3   0:01.93 systemd
    2 root        20   0    3180    2284   2072  S   0.0   0.1   0:00.00 init-systemd(Ub
    7 root        20   0    3196    2128    2080  S   0.0   0.1   0:00.00 init
   56 root       19  -1   50368   16292   15224  S   0.0   0.4   0:00.78 systemd-journal
  106 root        20   0   25156    6568    5148  S   0.0   0.2   0:01.18 systemd-udevd
  115 systemd+    20   0   21344   13224   11100  S   0.0   0.3   0:00.20 systemd-resolve
  116 systemd+    20   0   91036    7924    6952  S   0.0   0.2   0:00.23 systemd-timesyn
  168 root        20   0    4244    2688    2432  S   0.0   0.1   0:00.02 cron
  169 message+    20   0    9596    5416    4740  S   0.0   0.1   0:00.25 dbus-daemon
  183 root        20   0   17980    8848    7812  S   0.0   0.2   0:00.39 systemd-logind
  189 root        20   0  1755848   13264   10952  S   0.0   0.3   0:00.56 wsl-pro-service
  197 syslog      20   0   222516   5912    4528  S   0.0   0.2   0:00.23 rsyslogd
  209 root        20   0    3124    1976    1836  S   0.0   0.1   0:00.03 agetty
  229 root        20   0  107032   23024   13564  S   0.0   0.6   0:00.47 unattended-upgr
  295 root        20   0    3104    1108     980  S   0.0   0.0   0:00.00 SessionLeader
  312 root        20   0    3200    1244    1108  S   0.0   0.0   0:00.18 Relay(318)
  318 saireddy    20   0    6464    5664    3648  S   0.0   0.1   0:00.52 bash
  319 root        20   0    6704    4660    3876  S   0.0   0.1   0:00.06 login
  371 saireddy    20   0   20120   11320    9388  S   0.0   0.3   0:00.60 systemd
  374 saireddy    20   0   21168    3584    1856  S   0.0   0.1   0:00.00 (sd-pam)
  402 saireddy    20   0    6080    5380    3704  S   0.0   0.1   0:00.13 bash
  810 saireddy    20   0   9404    5484   3256  R   0.0   0.1   0:00.01 top
```